

Healthcare – Persistency of a Drug (Data Science)

Team Member's Details

Field	Information
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Batch Code	LISUM49
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Problem Description

The pharmaceutical industry continues to face major challenges in ensuring that patients remain adherent to prescribed therapies.

Non-adherence causes worse outcomes and higher healthcare costs.

ABC Pharma aims to predict and understand the factors influencing drug persistency.

Target variable:

- Persistency_Flag = 1 → Persistent
- Persistency_Flag = 0 → Non-persistent

The goal is to create a clean, model-ready dataset that supports accurate classification and interpretable insights for patient retention.

Data Overview

File: Healthcare_dataset.xlsx

Records: 3 424 **Columns:** 27 **Format:** Excel

Category	Example Features	Description
Demographics	Gender, Age, Ethnicity, Race, Region	Patient attributes

Physician Data	Ntm_Speciality, Ntm_Specialist_Flag	Prescriber information
Clinical/Risk	Risk_Segment_Prior_Ntm, Risk_Segment_During_Rx, T_Score_Change	Therapy and risk factors
Adherence Metrics	Adherent_Flag, Change_Risk_Segment	Behavior and compliance
Target	Persistency_Flag	Binary label

Data Inspection & Issues

Issue Type	Description	Example Columns
Implicit Missing Values	"Unknown" acts as NA	Race, Risk_Segment_Prior_Ntm
Outliers	Extreme values in durations/scores	Persistency_Duration, Risk_Score
Skewness	Right-skewed numeric distributions	Age, Risk_Score
Encoding Inconsistency	Mixed binary formats (Y/N vs Yes/No)	Adherent_Flag
Class Imbalance	Slight majority of non-persistent cases	Persistency_Flag

Missing-Values output

```

# Double check if there are any null (NA) values:

Null_Values = data.isnull().sum()

if Null_Values.any():
    print("Null values exist in the DataFrame.")
else:
    print("No null values exist in the DataFrame.")

```

[18]

```

... No null values exist in the DataFrame.

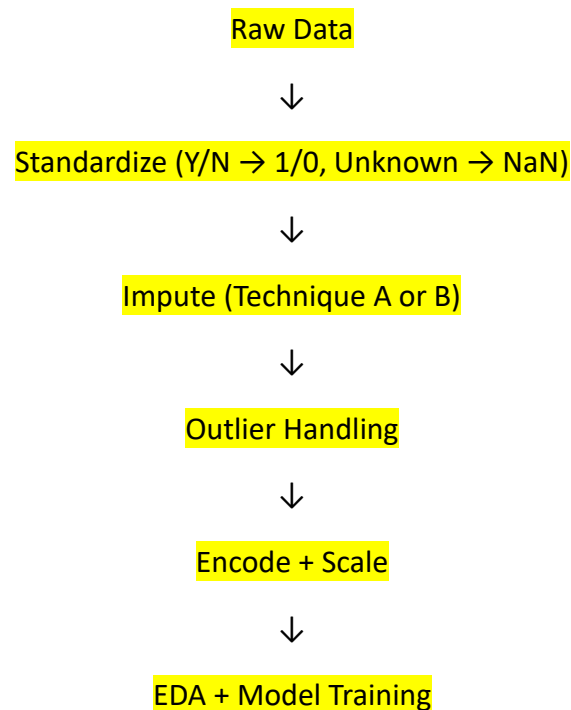
```

Data Cleansing and Transformation

Two distinct approaches were implemented as required.

Step	Technique A	Technique B
Missing Values	Replace “Unknown” → NaN → Impute with mode/median	KNN Imputer (model-based)
Outliers	IQR (Winsorization) – clip to $1.5 \times \text{IQR}$	Isolation Forest + Percentile Capping
Encoding	Binary → 1/0 ; OHE for categoricals	Same approach
Scaling	StandardScaler	StandardScaler

Workflow Pipeline



Rationale

- **Median/Mode Imputation** → simple and stable for small missing ratios.
- **KNN Imputation** → captures relationships across features.
- **IQR Capping** → robust against skew.
- **Isolation Forest** → detects non-linear outlier patterns.

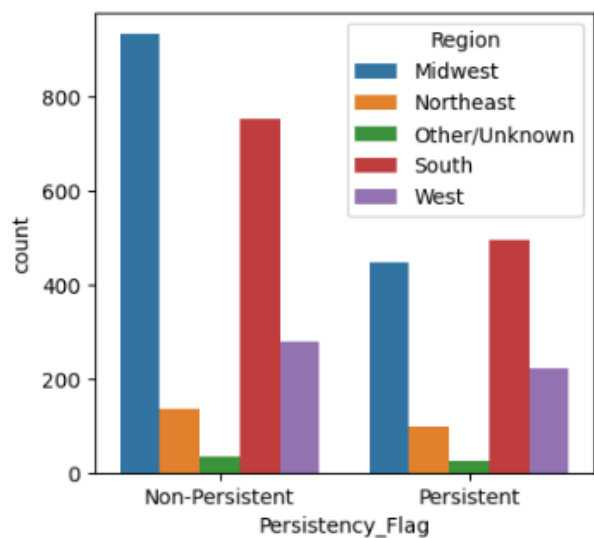
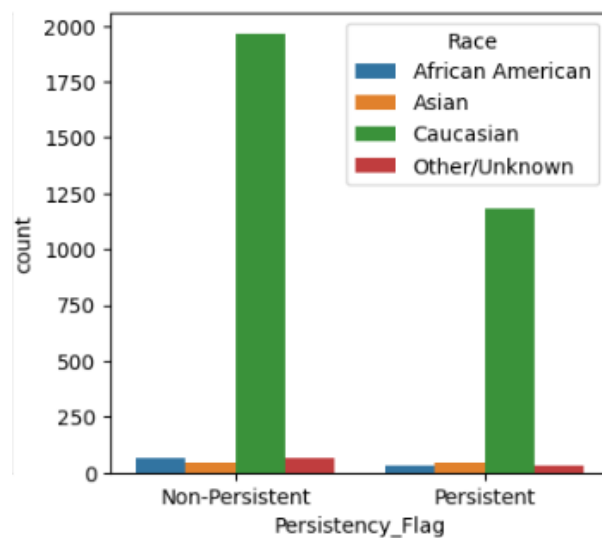
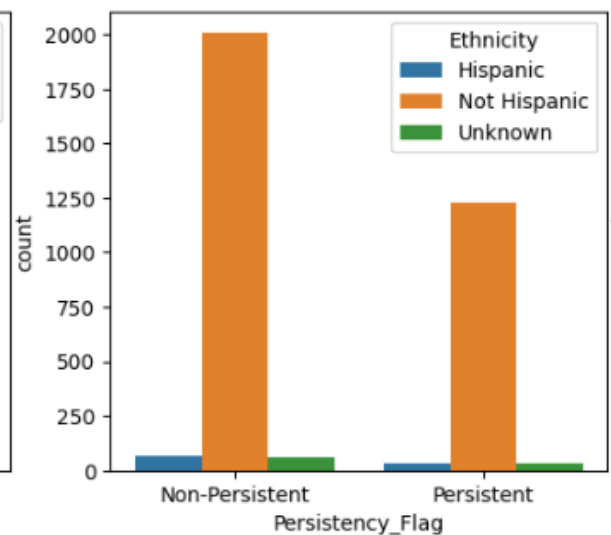
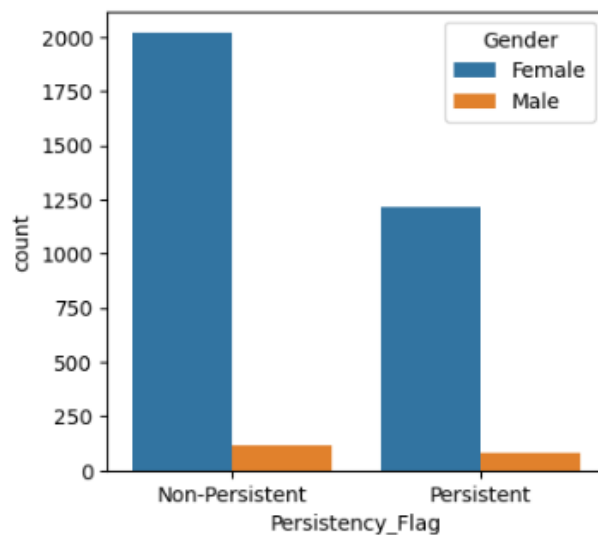
Exploratory Data Analysis (EDA)

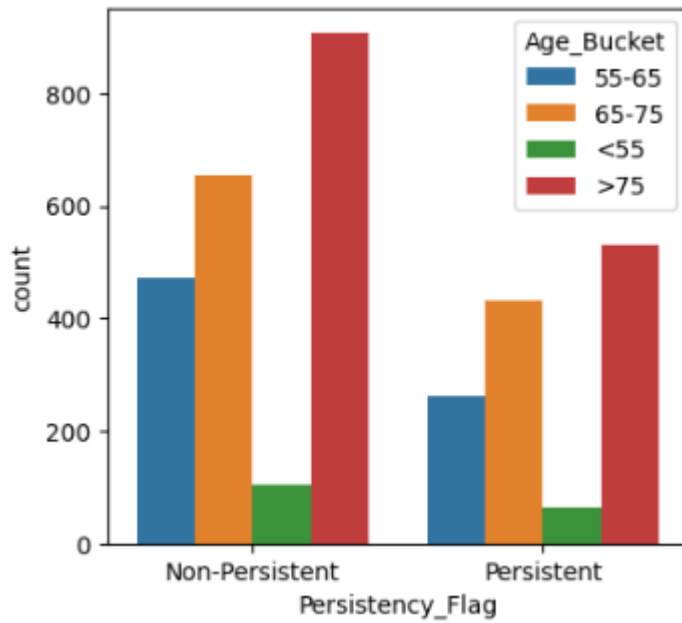
1. Target Distribution

55 % non-persistent vs 45 % persistent.

2. Demographic Insights

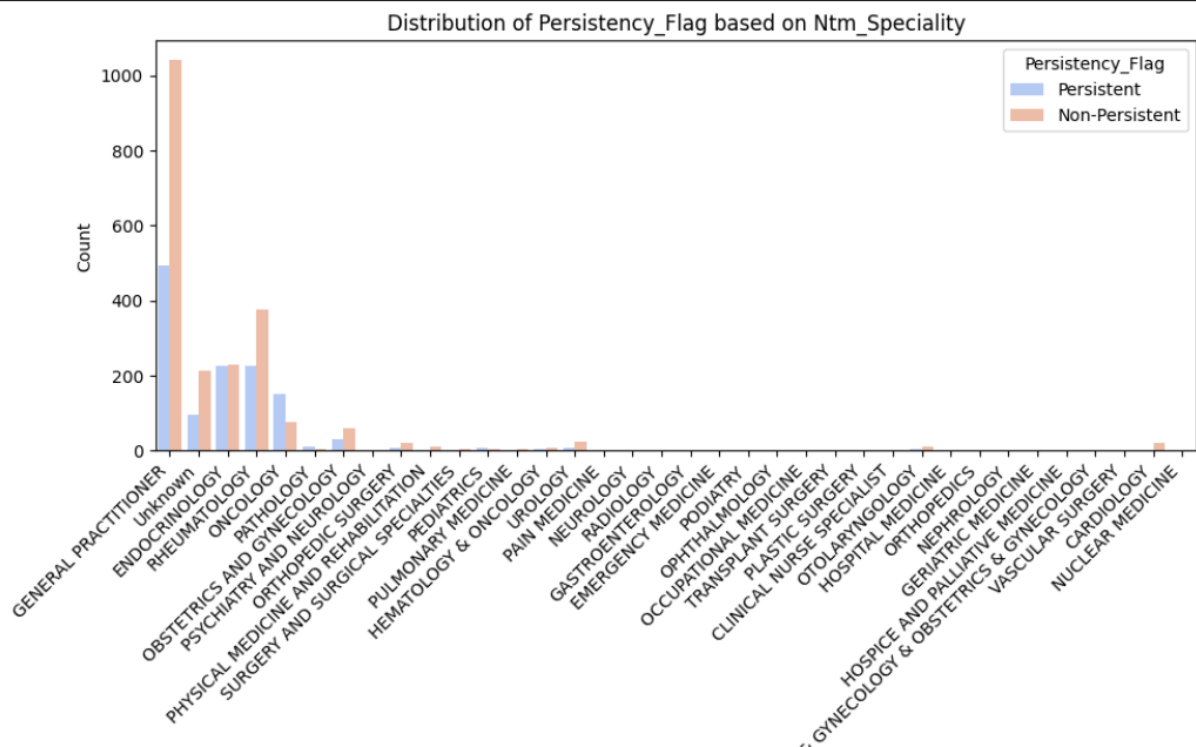
- **Gender:** Females > Males in both groups.
- **Ethnicity & Race:** Mostly Non-Hispanic Caucasian.
- **Region:** South and Midwest dominate.
- **Age:** Older patients show slightly higher persistency.

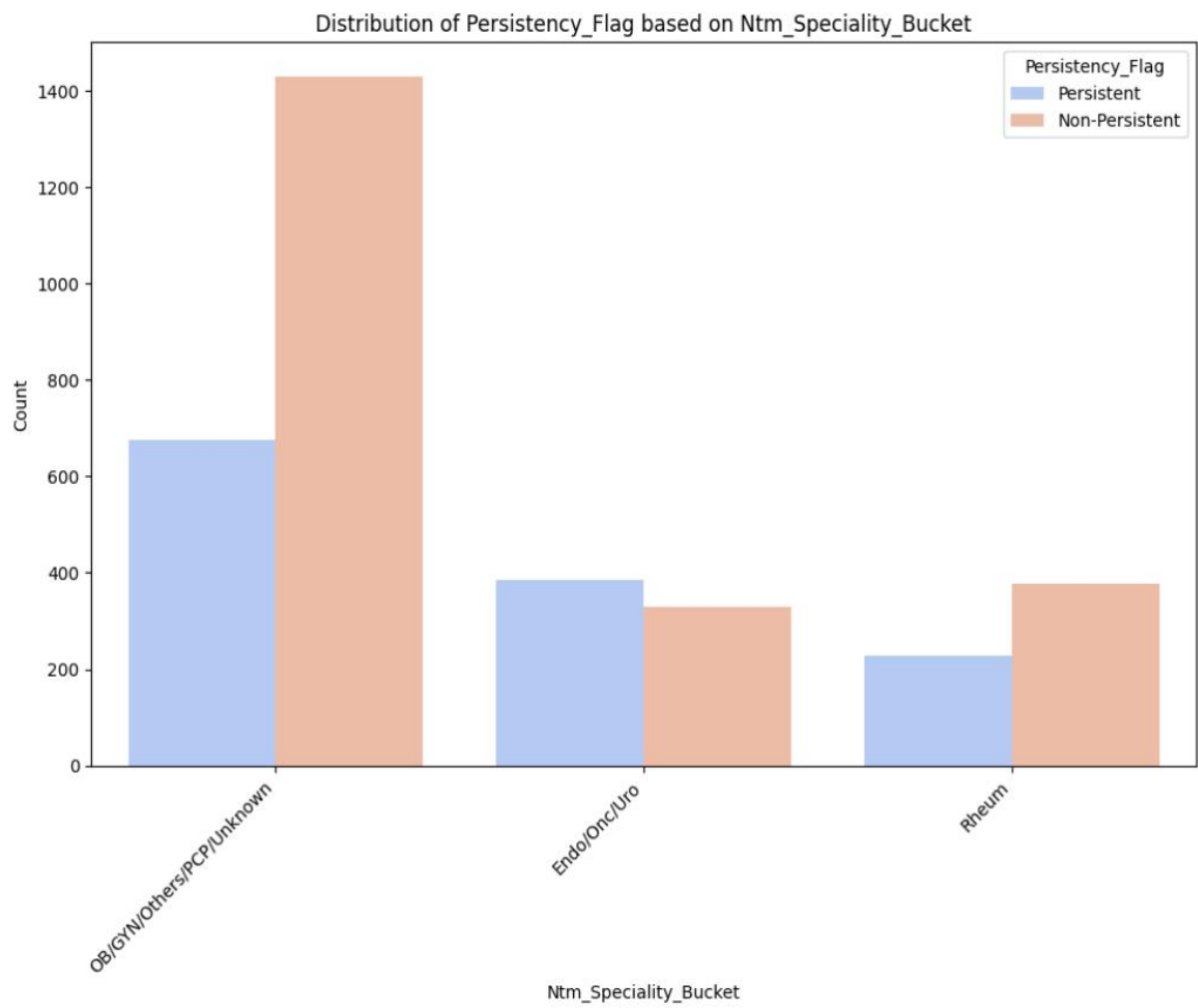
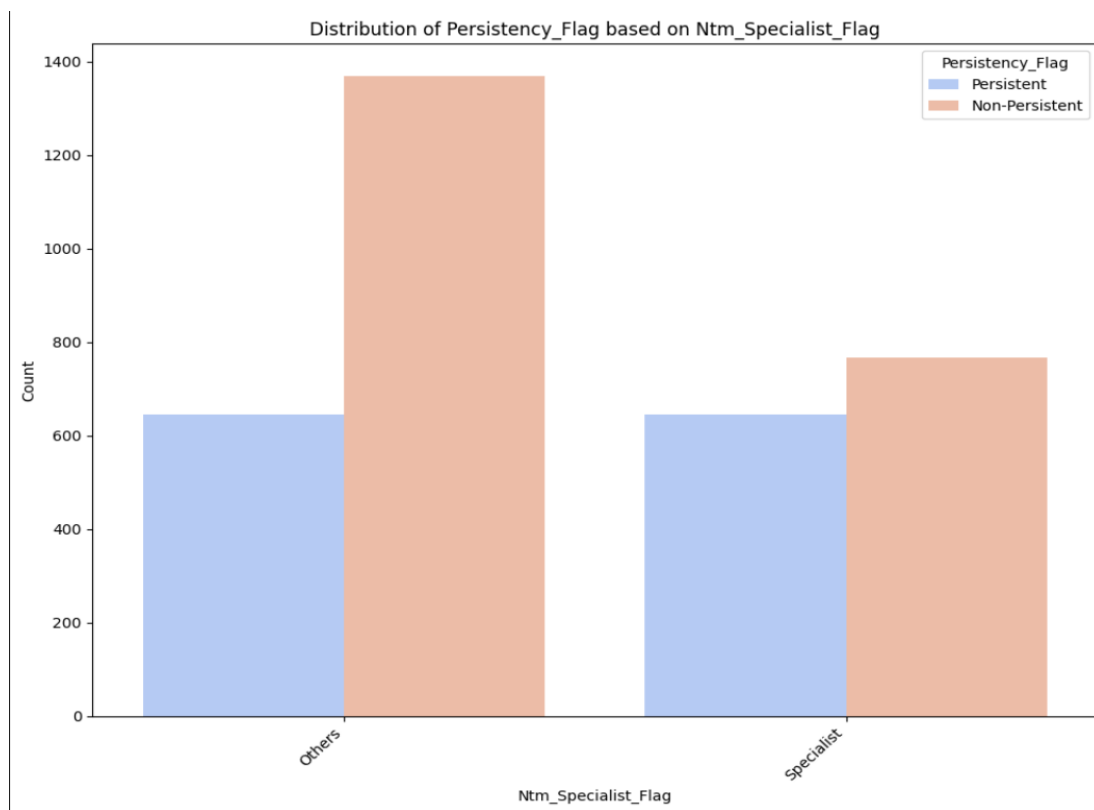




3. Provider Attributes

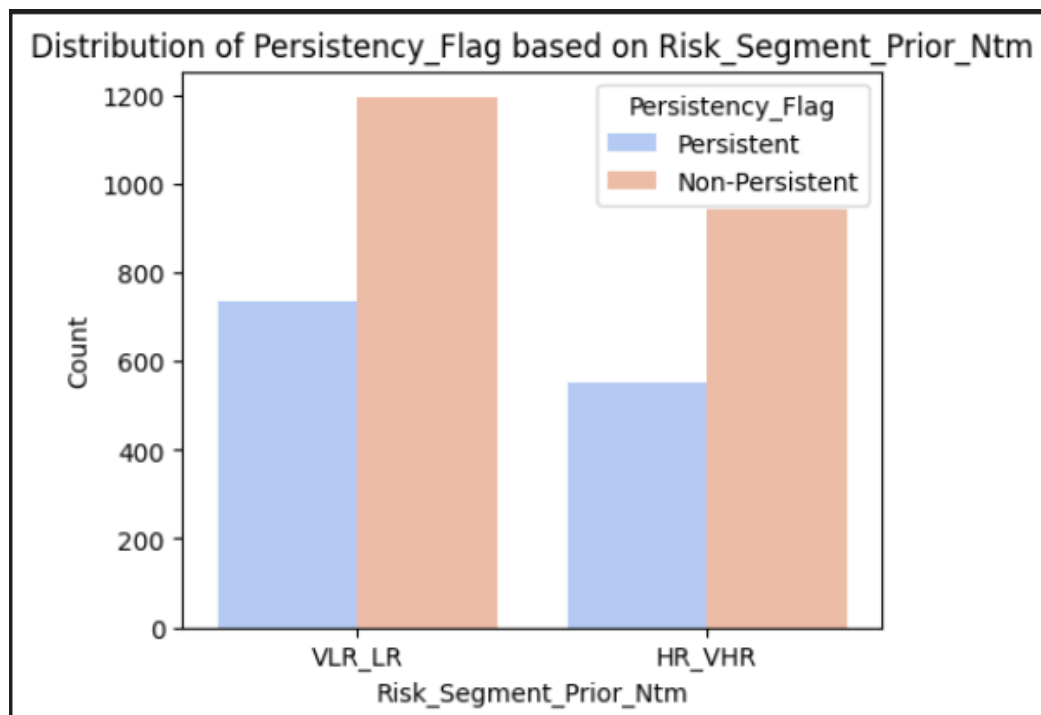
- General Practitioners and Rheumatologists account for most cases.
- Non-specialists → higher non-persistence.



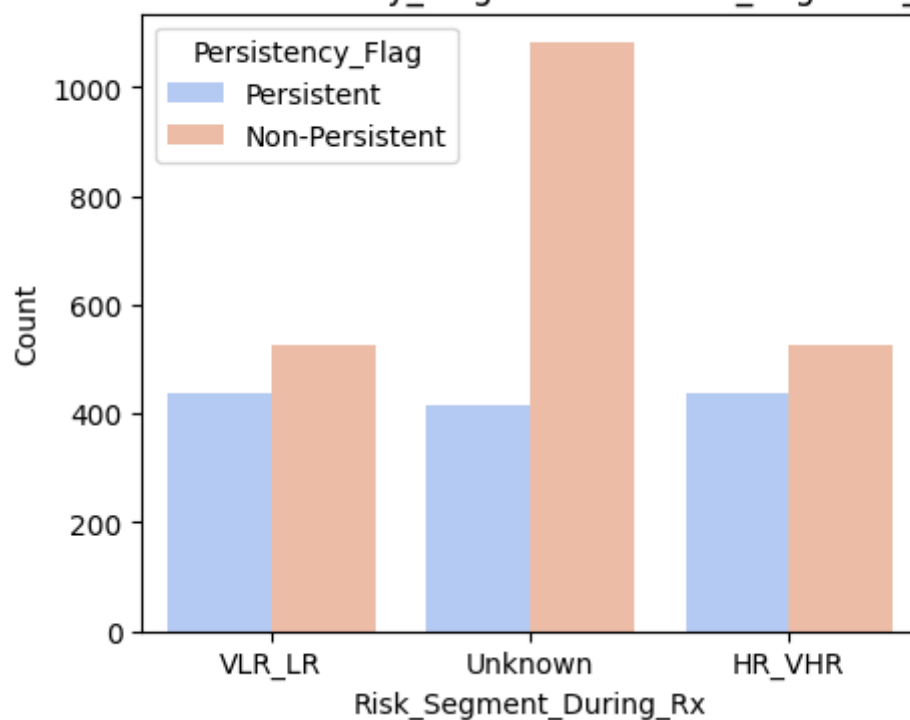


4. Risk & Clinical Variables

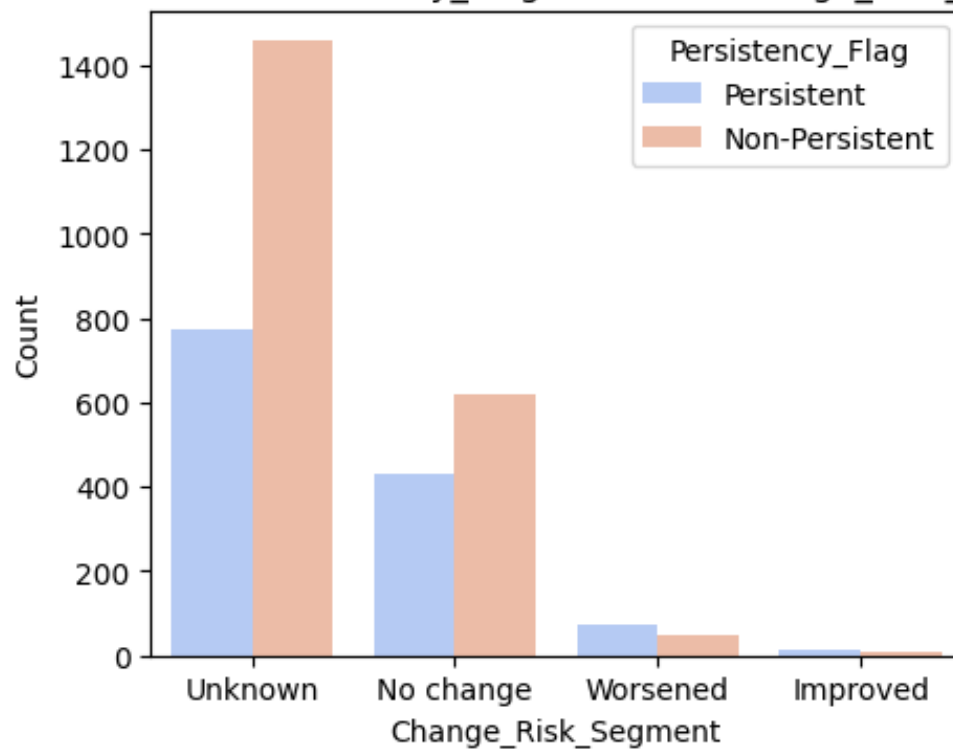
- Low-risk patients more likely to drop off.
- Minimal T-Score change in non-persistent patients.
- Adherent flag not a perfect indicator of persistency.



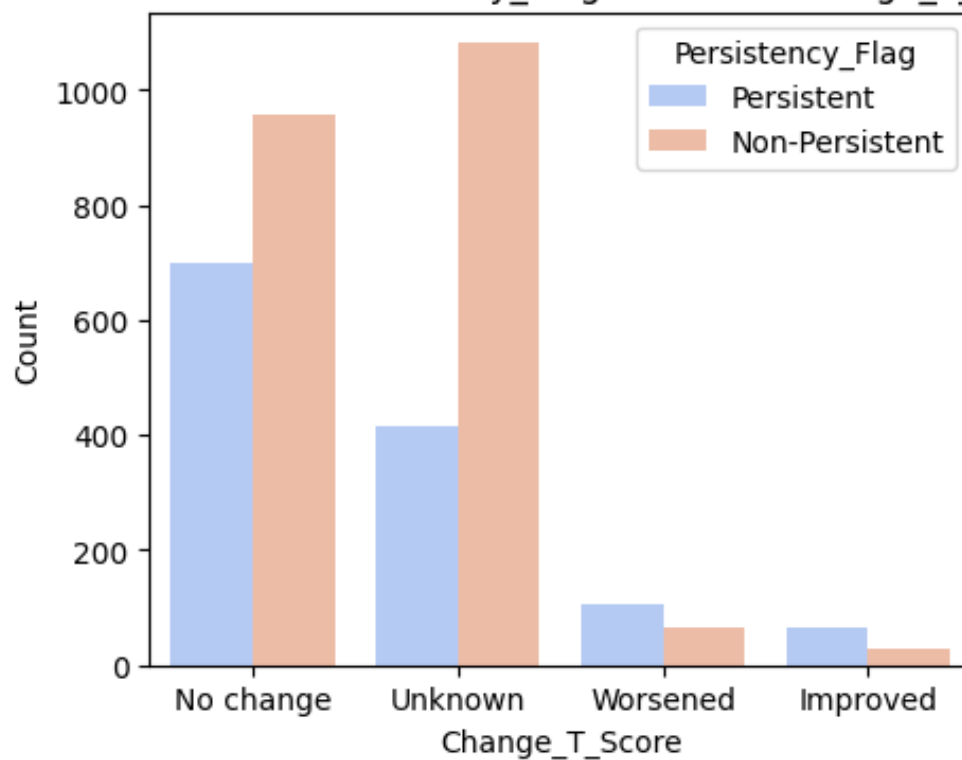
Distribution of Persistency_Flag based on Risk_Segment_During_Rx

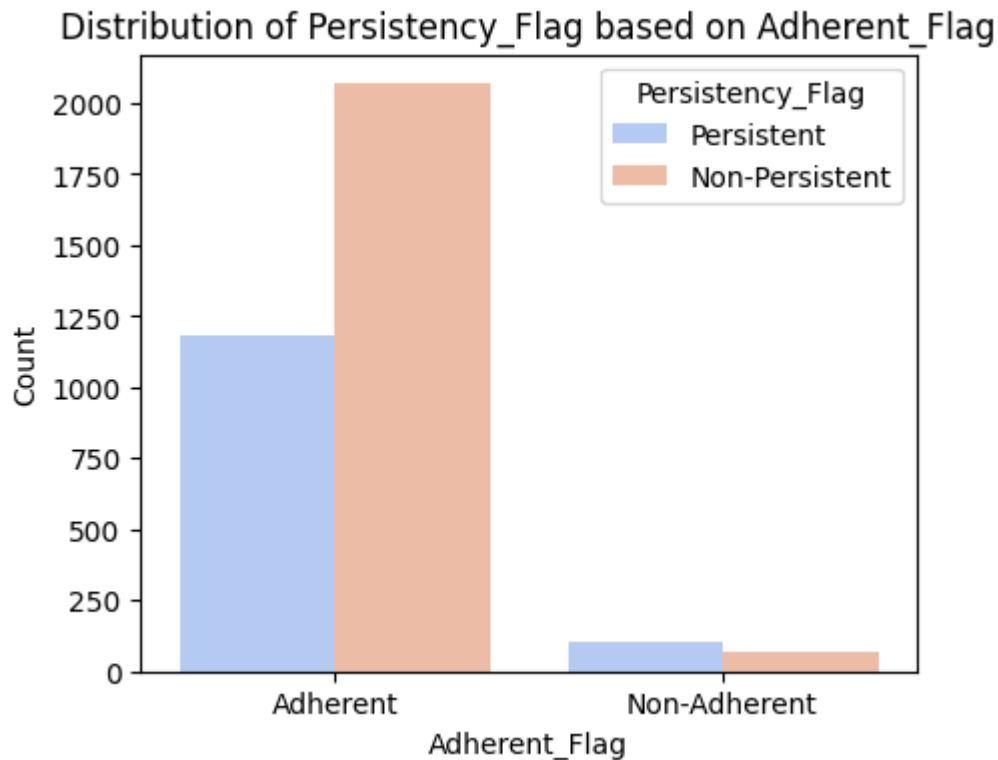


Distribution of Persistency_Flag based on Change_Risk_Segment



Distribution of Persistency_Flag based on Change_T_Score





Baseline Model & Results

Model: Logistic Regression

Split: 75/25 train-test

Technique	ROC-AUC	Comments
A (Median/Mode + IQR)	0.72	Good baseline stability
B (KNN + Isolation Forest)	0.75	Slight gain due to better outlier control

Review Process (GitHub)

- Solo project → self-review performed via GitHub PR comments.

Results & Conclusion

- **Best Pipeline:** KNN Imputer + Isolation Forest.
- **Most Influential Features:** Age, Risk Segment, Region.
- **Insight:** Patients with low initial risk and non-specialist physicians tend to drop earlier.

- **Next Steps:** Extend to tree-based models (Random Forest, XGBoost) and feature importance analysis.

GitHub Repository Link

<https://github.com/magdalena-ivanova-ds/Data-Glacier-Virtual-Internship/tree/main/week9>